Educator's Field Manual: The Inka Khipu Arcade & Somatic Compute System

A Complete Pedagogical Guide for Hands-On Andean Arithmetic

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1. Executive Summary & Pedagogical Foundations

Objective

To teach students positional decimal arithmetic, spatial matrix decomposition, and non-lossy digital logic without relying on abstract memorization, external data stacks, or electronic computation.

Core Philosophy: Somatic Computing & Discrete Phase Space

The **Yupana** (Andean counting board) and **Khipu** (knotted cord archive) form a unified, tactile information-processing architecture:

- **The Yupana is the ALU / RAM:** A dynamic, physical board where calculations occur in real time by moving physical counters (beans, corn kernels, or pebbles) directly within cup receptacles.
- **The Khipu is Non-Volatile Software Storage:** A permanent, physical record where final Yupana states are committed into cord networks using positional knot metrics.
- **Clock-Step Execution (Phase Space):** Instead of executing rigid mental math for every individual bean, the operator uses their hands to instantiate physical states by placing multiple beans in cups. At a chosen "clock step" (a deliberate manual pass across the board), the operator applies simple local rules (fusions, avalanches, unweaving) to let the system "settle" into its path of least resistance—its natural gravitational attractor.

Key Learning Outcomes

By completing this module, students will:

1. Express any decimal number 0...9,999 on a physical 4-cup geometric grid using no more than 3 physical counters per row.
2. Perform addition and subtraction using fluid "fusions," "avalanches," and "unweaving" instead of memorized carry/borrow algorithms.
3. Execute multi-digit multiplication and division entirely on-board through spatial loom shifts and fair-share allocation (*Ayni*).
4. Understand how tactile, non-silicon systems process complex computational workloads deterministically.

2. Materials & Physical Set-Up

Classroom Materials Required (Per Student or Pair)

1. **1 × Yupana Board:** A grid of 5 rows by 4 geometric columns.
    - *Classroom Alternatives:* Readily available egg cartons (a 5×4 grid of egg cups) or drawn grids on paper/whiteboards.
2. **30–40 Physical Counters:** Quinoa beans, dry corn kernels, small river stones, or plastic tokens. *(No cards, paper stacks, or external state trackers required.)*

The Board Topology

TIER / ROW	[5]	[3]	[2]	[1]	TIER VALUE
10^4 **(10,000)**	( )	( )	( )	( )	= 0
10^3 **(1,000)**	( )	( )	( )	( )	= 0
10^2 **(100)**	(●)	( )	(●)	( )	= **700** *[(5 + 2) × 100]*
10^1 **(10)**	( )	(●)	( )	(●)	= **40** *[(3 + 1) × 10]*
10^0 **(1)**	(●)	(●)	( )	(●)	= **9** *[(5 + 3 + 1) × 1]*

- **Vertical Axis (Powers of 10):** Bottom row is Units (10^0), moving up to Tens (10^1), Hundreds (10^2), Thousands (10^3), and Ten-Thousands (10^4).
- **Horizontal Axis (The 4 Magic Cups):** Each row contains four physical receptacles valued at **[5]**, **[3]**, **[2]**, and **[1]**.

3. Unit 1: Digit Representation & The Golden Rule

Lesson Concept

Any decimal digit ($0 \dots 9$) is formed in a single row by summing the values of the cups containing beans.
The Golden Rule: Always place counters in the largest available cup that fits the remaining value.
This rule guarantees that **no single digit ever requires more than 3 beans** when the board is in a rested state.

Teacher's Digit Encoding Reference Chart

Digit	Cup Combination	Total Beans Used
0	Empty Row	0
1	[1]	1
2	[2]	1
3	[3]	1
4	[3] + [1]	2
5	[5]	1
6	[5] + [1]	2
7	[5] + [2]	2
8	[5] + [3]	2
9	[5] + [3] + [1]	3

4. Unit 2: Somatic Addition & The Clock-Step Avalanche

Mechanical Principles: Phase Space & Clock Steps

In somatic compute, addition is performed by directly placing the new addend's beans into the board alongside the existing value. You do not need to calculate carries immediately while placing beans. The human operator populates the cups (instantiating the un-settled phase space) and then "steps the clock" forward by sweeping hands across the board to apply local settlement rules:

1. **In-Row Fusion (Gravitational Attraction):**
    - **Two [1] beans in cup [1]:** On clock step, merge into one bean in cup [2].
    - **One [1] and one [2] bean:** On clock step, merge into one bean in cup [3].
    - **One [2] and one [3] bean:** On clock step, fall into the path of least resistance → merge into one bean in cup [5]. Note, there is a “gravity” to the golden rule’s movement, which will become intuitive with practice. We could represent a 5 by populating 3 and 2, but the clock step essentially “applies gravity and fusion” at time of calculation.
2. **The Great Bean Avalanche (Carry-Up Rule):**
    - When cup [5] accumulates **two beans** ($5 + 5 = 10$), both beans fuse and **jump UP to the tier above as a single [1] bean** ($+ 10$). Cup [5] of the current tier is cleared to 0.

TIER	**[5]**	**[3]**	**[2]**	**[1]**
Tier Above (10s)	( )	( )	( )	**(●)** *(Carried +1 bean into Cup [1])*
Avalanche Trigger	\	\	\	**▲ (5 + 5 = 10 Avalanche Trigger)**
Units Tier (1s)	**(●) + (●)**	( )	( )	**Cup [5] empties to 0**

Direct On-Board Addition Procedure (A + B)

1. Set the initial number A on the Yupana according to the Golden Rule.
2. Take the beans for number B in your hand.
3. Drop B's beans directly into the corresponding tier cups where A is sitting (e.g., if B has a 5 in the units place, drop a bean into the units row's [5] cup).
4. **Clock Step:** Sweep your hands from bottom tier (10^0) upwards, resolving fusions and avalanches until every row obeys the Golden Rule.

5. Unit 3: Somatic Subtraction & Unweaving (Borrowing)

Mechanical Principle

Subtraction is the physical removal of beans directly from the board. If a cup does not contain enough beans to remove the subtrahend value, the operator **unweaves** a bean from a higher tier to supply the necessary energy.

Unweaving Rule (Borrowing): Remove 1 bean from cup [1] of the tier above (10^{n+1}), and replace it with TWO beans in cup [5] of the current tier (10^n). ($10 = 5 + 5$).

TIER	ACTION / BEAN STATE
Tier Above (10s)	**(●)** → *Remove 1 bean (-10)*
Unweaving Flow	**| → ▼ (Unweaves into 5 + 5)**
Units Tier (1s)	**(●) (●) in Cup [5]** → *Subtract freely!*

Direct Subtraction Procedure (A - B)

1. Set minuend A on the Yupana.
2. Identify the value of subtrahend B by digit.
3. Remove beans corresponding to B starting from the highest tier down. If a cup lacks a required bean, unweave one bean from the tier above into two [5] beans in the current tier, then remove the target amount.

6. Unit 4: Tactile Multiplication (The Multiplier Loom)

Mechanical Principle

Multiplication on the Yupana uses spatial translation across rows—treating the vertical board tiers as "loom rails." You do not need to memorize times tables.

Spatial Shift Rules

Decompose the multiplier into its individual component cups ($[5]$, $[3]$, $[2]$, $[1]$ across their respective rows). For each component:

- **Multiplying by [1]:** Replicate the multiplicand's pattern in the same row tier.
- **Multiplying by [2]:** Double the multiplicand's pattern into the same row tier.
- **Multiplying by [3]:** Triple the multiplicand's pattern into the same row tier.
- **Multiplying by [5]:** Shift the multiplicand's pattern UP one row tier (equivalent to $\times\ 10$) and take half, OR place into the [5] cups of the shifted tier.
- **Multiplying by Tens / Hundreds (10, 30, etc.):** Shift the entire pattern UP by 1 or 2 row tiers.

After placing all partial product beans directly onto the board, execute a single **Clock Step** to resolve all fusions and avalanches until the board settles.

7. Unit 5: Somatic Division & Fair-Share Allocation (*Ayni*)

Pedagogical Foundations of *Ayni*

In Andean culture, *Ayni* represents reciprocal community exchange and fair-share distribution. Somatic division models this directly: a total communal harvest (dividend) is systematically partitioned into equal family allotments (divisor).

Division on the Yupana is the **exact inverse of the Multiplier Loom**. Instead of shifting a multiplicand UP the board to build a product, division tests how many full divisor units can be "harvested" (subtracted) from the dividend, tier by tier, working from the top down.

The 4-Step Somatic Division Protocol

STEP	ACTION	DESCRIPTION
1	**Board Setup**	Place the total dividend on the main Yupana grid. Keep the divisor to the side as a reference template.
2	**Test Capacity**	Test if a full unit of the divisor (scaled to the highest occupied tier) fits into the dividend.
3	**Harvest & Allot**	Subtract the divisor's pattern from the board and record 1 count in the Quotient row for that tier.
4	**Step-Down**	Unweave remaining beans down ($10^{n} \rightarrow 5 + 5$ in 10^{n-1}) and repeat until Units are complete.

Detailed Division Walkthrough 1: 4,800 / 32 = 150

- **Initial Board State:** 4,800 on board (Thousands: [3]+[1], Hundreds: [5]+[3]). Divisor reference = 32 (3 Tens, 2 Units).

Step 1: Test Hundreds Tier (Scaling Divisor by 100 → 3,200)

- Can we remove 3,200 (3 Thousands, 2 Hundreds) from 4,800? **YES!**
- **Action:** Remove 3 beans from Thousands row, 2 beans from Hundreds row.
- **Board Remaining:** 1,600 (Thousands: [1], Hundreds: [5]+[1]).
- **Quotient Recorded:** 1 Hundred (100).

Step 2: Test Hundreds Tier Again

- Can we remove another 3,200 from 1,600? **NO.**
- **Unweave Thousands:** Take 1 bean from Thousands [1] and unweave it into TWO [5] beans in Hundreds row.
- **Board Hundreds Row now holds:** [5], [5], [5], [1] (= 1,600).

Step 3: Test Tens Tier (Scaling Divisor by 10 → 320)

- How many 320s (3 Hundreds, 2 Tens) fit into 1,600?
    - 1st allotment (-320): Board = 1,280. Quotient += 10.
    - 2nd allotment (-320): Board = 960. Quotient += 10.
    - 3rd allotment (-320): Board = 640. Quotient += 10.
    - 4th allotment (-320): Board = 320. Quotient += 10.
    - 5th allotment (-320): Board = 0. Quotient += 10.
- **Total Tens Allotment:** 50.

FINAL RESULT: $100 + 50 = 150$ *(Exact Distribution, 0 Remainder)*

Detailed Division Walkthrough 2: 1,472 / 23 = 64

- **Initial Board State:** 1,472 on board (Thousands: [1], Hundreds: [3]+[1], Tens: [5]+[2], Units: [2]). Divisor reference = 23 (2 Tens, 3 Units).

Step 1: Test Hundreds Tier (Scaling Divisor by 100 → 2,300)

- Can we remove 2,300 from 1,472? **NO.**
- **Unweave Thousands:** Take 1 bean from Thousands [1] and unweave into TWO [5] beans in Hundreds row.
- **Hundreds row now holds:** 1,400 (two [5]s, one [3], one [1]).

Step 2: Test Tens Tier (Scaling Divisor by 10 → 230)

- How many 230s (2 Hundreds, 3 Tens) fit into 1,472?
    - Test 60 allotments ($6 \times 230 = 1,380$):
    - Subtract 1,380 (13 Hundreds, 8 Tens) from board.
    - **Board Remaining:** 92 (Tens: [5]+[3]+[1], Units: [2]).

- **Quotient Recorded:** 60.

Step 3: Test Units Tier (Divisor → 23)

- How many 23s (2 Tens, 3 Units) fit into 92?
  - Unweave Tens if needed to expose unit beans.
  - 4 allotments ($4 \times 23 = 92$).
  - Subtract 92 from board. **Board Remaining:** 0.
  - **Quotient Recorded:** 4.

FINAL RESULT: $60 + 4 = 64$ *(Exact Distribution)*

8. Unit 6: The Khipu Archive (Database Commits)

Once a calculation settles on the Yupana, the state is committed into a **Khipu pendant cord**:

PRIMARY CORD (MAIN RAIL)	**Pendant 1**	**Pendant 2**	**Pendant 3**
Thousands (10^3)	(S)	( )	(L3) 3,000
Hundreds (10^2)	(L7)	(S) Tens	( )
Units (10^0)	(E)	(L4)	(E)
COMMITTED VALUE	**1,701**	**14**	**3,001**

Knot Metric Encoding

1. **Single Knots (S):** Used in Thousands (10^3) tier and higher.
2. **Long Knots (L):** Used in Tens (10^1) and Hundreds (10^2) tiers. The number of finger-turns in the knot equals the digit (e.g., $L7 = 7$).
3. **Figure-Eight Knots (E):** Used exclusively in the Units (10^0) tier to represent a 1.
4. **Empty Cord Space:** Represents a 0 placeholder for that positional tier.

Note: The tying of the knot types is not something we have enough skill to teach effectively, but we are decoding with these baselines. Supplying thread to the students may allow them to discover this system on their own, as it is a discoverable method of storing software. There are colors involved in some of the later khipu encoding schemes, as well, and this system does not

exhaust the complete compute ecosystem of the Andean khipu tradition. There is evidence of more advanced Yupana board configurations, perhaps used to track advanced or remote state. There are also items that appear to be something akin to smart contracts. We will continue publishing and distributing materials as we learn more.

9. Educator's Assessment & Verification Rubric

Skill / Objective	Novice (1-2 pts)	Proficient (3-4 pts)	Master (5 pts)
Digit Encoding	Uses > 3 beans per row or violates the Golden Rule.	Encodes digits $0\dots9$ correctly using the Golden Rule.	Instantly sets any 4-digit number with minimal bean counts.
Addition & Clock Step	Forgets to fuse beans or misses carry-ups across rows during the clock step.	Executes In-Row Fusions and Bean Avalanches accurately on board.	Demonstrates fluid multi-row avalanches without prompting.
Subtraction & Unweaving	Attempts to subtract from empty cups without unweaving higher tiers.	Unweaves higher-tier beans ($10 \to 5 + 5$) correctly.	Performs multi-tier unweaving smoothly during complex problems.
Ayni Division	Struggles to scale divisor tiers or track subtractive allotments.	Executes top-down harvesting and unweaving to find exact quotients.	Demonstrates fluid division and explains *Ayni* resource allocation.
Database Commit	Confuses knot types or positional tiers on the pendant cord.	Accurately translates Yupana values onto Khipu cord metrics (s, L, E).	Explains how Yupana state transfers non-lossy into Khipu storage.

10. Updated Sample Student Worksheets

Worksheet 1: Digit Encoding Practice

Mark or place the beans in the cups for each number according to the Golden Rule:

1. **Number: 382**
    - Hundreds (10^2): [5] ( ) [3] (●) [2] ( ) [1] ( )
    - Tens (10^1): [5] (●) [3] (●) [2] ( ) [1] ( )
    - Units (10^0): [5] ( ) [3] ( ) [2] (●) [1] ( )
2. **Number: 1,509**
    - Thousands (10^3): [5] ( ) [3] ( ) [2] ( ) [1] (●)
    - Hundreds (10^2): [5] (●) [3] ( ) [2] ( ) [1] ( )
    - Tens (10^1): [5] ( ) [3] ( ) [2] ( ) [1] ( ) *(Empty)*
    - Units (10^0): [5] (●) [3] (●) [2] ( ) [1] (●)

Worksheet 2: Somatic Problem Set (On-Board Operations)

Solve using your physical Yupana board (no external stacks!). Record your intermediate steps and final board state:

Problem 1: Addition

$247 + 385 =$ _____

- *Step A:* Set 247 on board.
- *Step B:* Drop 385 directly into corresponding tier cups.
- *Step C (Clock Step):* Resolve fusions and avalanches.
- **Target Answer: 632** *(Hundreds: [5]+[1], Tens: [3], Units: [2])*

Problem 2: Subtraction

$1,204 - 531 =$ _____

- *Step A:* Set 1,204 on board.
- *Step B:* Unweave Thousands tier ($1,000 \rightarrow 500 + 500$ in Hundreds).
- *Step C:* Remove 531 directly.
- **Target Answer: 673** *(Hundreds: [5]+[1], Tens: [5]+[2], Units: [3])*

Problem 3: Multiplication (Loom Shift)

$64 \times 23 =$ _____

- *Step A:* Decompose multiplier $23 = 20 + 3$.
- *Step B:* Place 64×2 shifted up 1 row ($1,280$).
- *Step C:* Place 64×3 in same row (192).
- *Step D (Clock Step):* Resolve all fusions/avalanches.
- **Target Answer: 1,472** *(Thousands: [1], Hundreds: [3]+[1], Tens: [5]+[2], Units: [2])*

Problem 4: Ayni Division (Fair-Share Allocation)

$3,600/24 =$ _____

- *Step A:* Set 3,600 on board. Divisor = 24.
- *Step B (Hundreds Tier):* Test 240s (24×10). Subtract $2,400$ (1×2400). Remaining = $1,200$. Quotient Hundreds = 100.
- *Step C (Tens Tier):* Unweave $1,200$ into Tens/Hundreds. Test 24s (24×10). Subtract $1,200$ (5×240). Remaining = 0. Quotient Tens = 50.
- **Target Answer: 150** *(Quotient: Hundreds [1], Tens [5])*